Q5) Write a C program for generalization of the Caesar cipher, known as the affine Caesar cipher, has the following form: For each plaintext letter p, substitute the ciphertext letter C: C = E([a, b], p) = (ap + b) mod 26 A basic requirement of any encryption algorithm is that it be one-to-one. That is, if p q, then E(k, p) E(k, q). Otherwise, decryption is impossible, because more than one plaintext character maps into the same ciphertext character. The affine Caesar cipher is not one-to-one for all values of a. For example, for a = 2 and b = 3, then E([a, b], 0) = E([a, b], 13) = 3.

a. Are there any limitations on the value of b?

b. Determine which values of a are not allowed.

Aim: To write a C program to implement the **Affine Caesar Cipher** using the encryption formula **C = (aP + b) mod 26**, and determine the valid values of **a** and **b** for successful encryption and decryption.

Algorithm:

1. Start the program.
2. Declare variables for plaintext, ciphertext, keys **a** and **b**, and loop counter.
3. Read the plaintext, value of **a**, and value of **b** from the user.
4. Check whether **a** is coprime with 26 (i.e., GCD(a,26) = 1). If not, display an error message and terminate the program.
5. Traverse each character of the plaintext.
6. Convert each alphabetic character into its numerical equivalent (A = 0, B = 1, ..., Z = 25).
7. Encrypt each character using the formula: **C = (a × P + b) mod 26**.
8. Convert the encrypted numerical value back to its corresponding alphabet and store it in the ciphertext.
9. Display the encrypted ciphertext.
10. Stop the program..

Result:

Thus, the C program to implement the **Affine Caesar Cipher** was successfully executed. It was observed that **b** can take any value from **0 to 25**, whereas **a** must be **coprime with 26** to ensure a unique and reversible encryption process.

Answer to the Questions

a. Are there any limitations on the value of b?

Answer:

No. The value of **b** can be **any integer from 0 to 25** because it simply shifts the encrypted letters after multiplication. It does not affect whether the cipher is reversible.

b. Determine which values of a are not allowed.

Answer:

The value of **a** must be **coprime with 26**, i.e., **GCD(a, 26) = 1**.

The **allowed values** of **a** are:

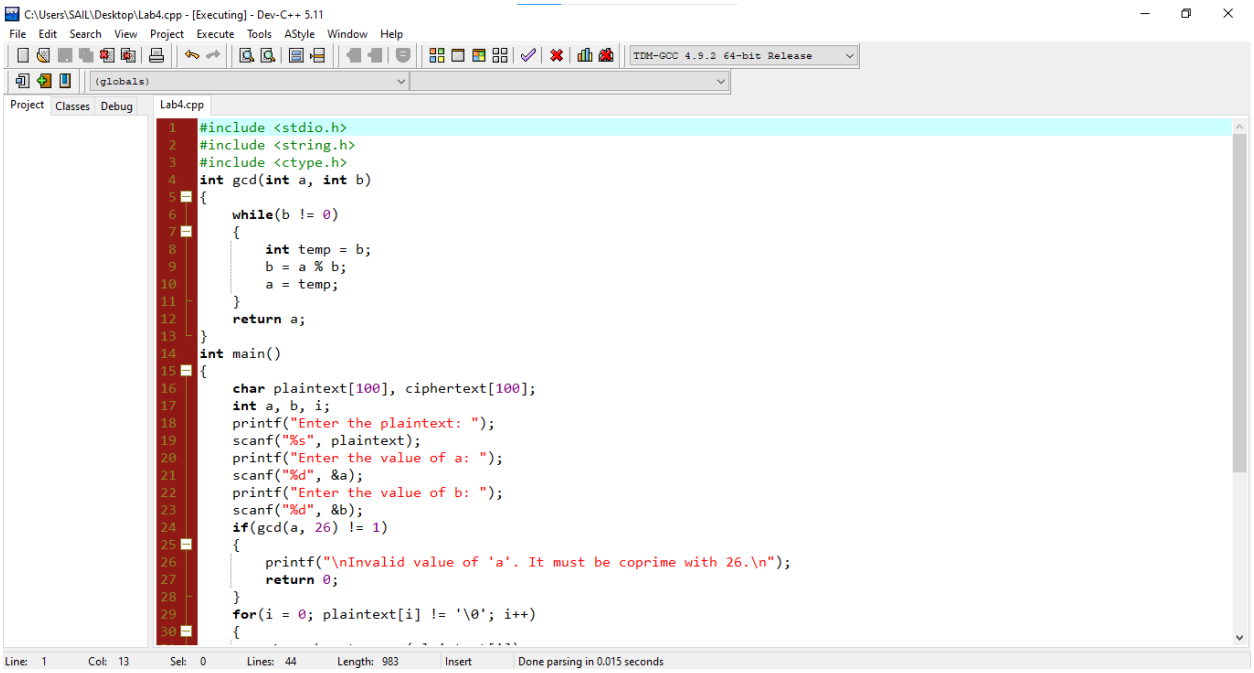
1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

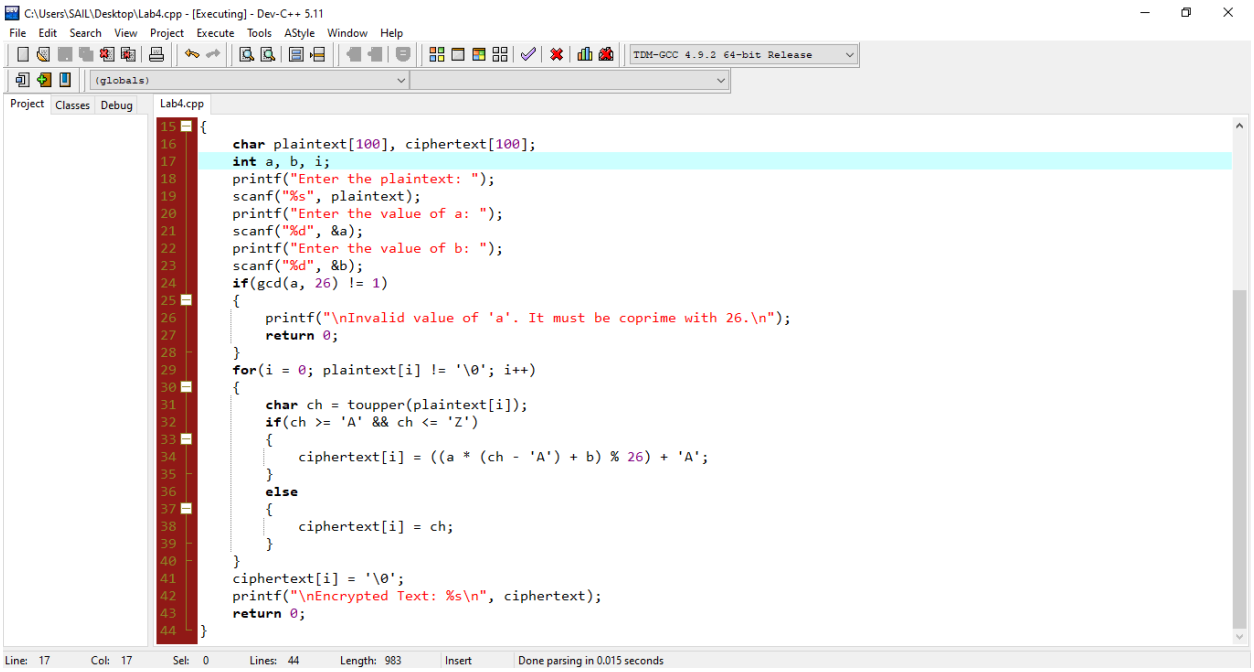
The **not allowed values** are:

0, 2, 4, 6, 8, 10, 12, 13, 14, 16, 18, 20, 22, 24

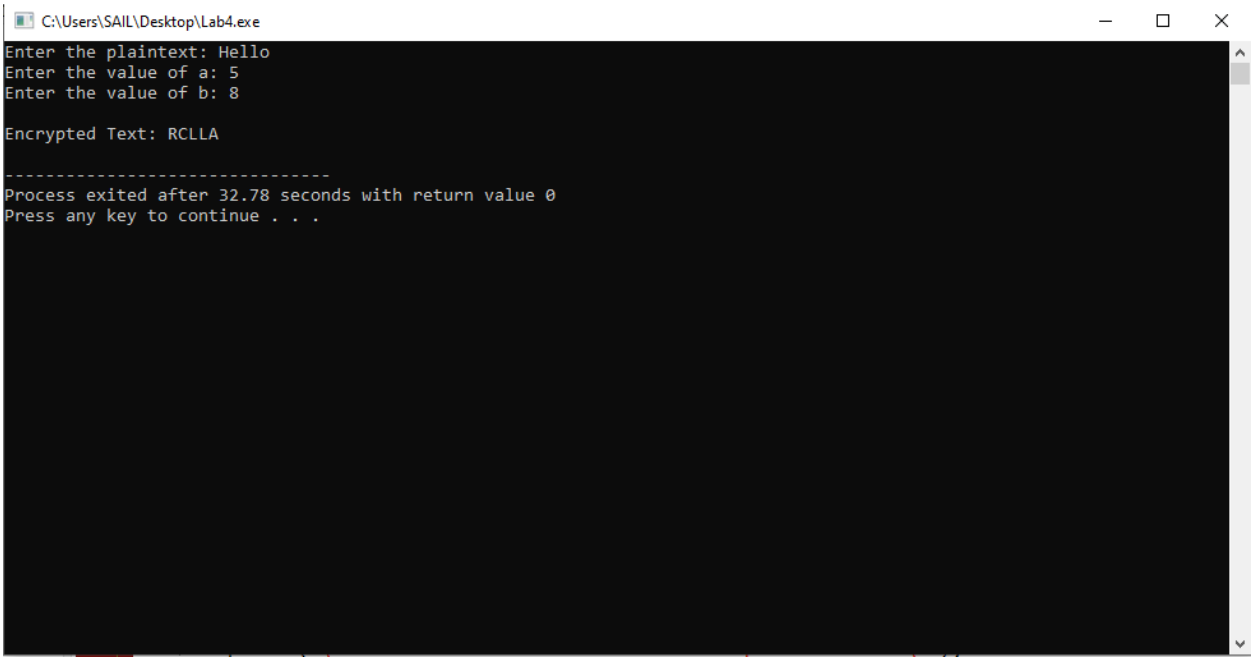
These values share a common factor with 26, making the encryption **not one-to-one**, so decryption becomes impossible.

Code:


C:\Users\SAIL\Desktop\Lab4.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
Lab4.cpp
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int gcd(int a, int b)
{
while(b != 0)
{
int temp = b;
b = a % b;
a = temp;
}
return a;
}
int main()
{
char plaintext[100], ciphertext[100];
int a, b, i;
printf("Enter the plaintext: ");
scanf("%s", plaintext);
printf("Enter the value of a: ");
scanf("%d", &a);
printf("Enter the value of b: ");
scanf("%d", &b);
if(gcd(a, 26) != 1)
{
printf("\nInvalid value of 'a'. It must be coprime with 26.\n");
return 0;
}
for(i = 0; plaintext[i] != '\0'; i++)
{
Line: 1 Col: 13 Sel: 0 Lines: 44 Length: 983 Insert Done parsing in 0.015 seconds


C:\Users\SAIL\Desktop\Lab4.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
Lab4.cpp
{
char plaintext[100], ciphertext[100];
int a, b, i;
printf("Enter the plaintext: ");
scanf("%s", plaintext);
printf("Enter the value of a: ");
scanf("%d", &a);
printf("Enter the value of b: ");
scanf("%d", &b);
if(gcd(a, 26) != 1)
{
printf("\nInvalid value of 'a'. It must be coprime with 26.\n");
return 0;
}
for(i = 0; plaintext[i] != '\0'; i++)
{
char ch = toupper(plaintext[i]);
if(ch >= 'A' && ch <= 'Z')
{
ciphertext[i] = ((a * (ch - 'A') + b) % 26) + 'A';
}
else
{
ciphertext[i] = ch;
}
}
ciphertext[i] = '\0';
printf("\nEncrypted Text: %s\n", ciphertext);
return 0;
}
Line: 17 Col: 17 Sel: 0 Lines: 44 Length: 983 Insert Done parsing in 0.015 seconds

Output:


C:\Users\SAIL\Desktop\Lab4.exe
Enter the plaintext: Hello
Enter the value of a: 5
Enter the value of b: 8
Encrypted Text: RCLLA

Process exited after 32.78 seconds with return value 0
Press any key to continue . . .